

test_data.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/atm/test_data.vr

```
#include "atm.vrh"
program test_data
{
    rvm_log log = new("Data Test", "Program");
    integer i = 0;
    repeat (100) {
        atm_cell cell = new;
        atm_cell cpy;
        bit [7:0] bytes[*];
        string diff;
        integer n, m;
        integer fp;
        if (!cell.randomize()) {
            rvm_error(log, "Cell did not succesfully randomize");
        }

        // Does it display right?
        cell.display("");
        if (!cell.is_valid(0)) {
            rvm_error(log, "Cell did not check as valid");
        }
        if (!cell.compare(cell, diff)) {
            rvm_error(log, "Cell did not compare to self");
        }

        rvm_trace(log, "Copying frame to new instance...");
        cast_assign(cpy, cell.copy());
        if (!cpy.compare(cell, diff)) {
            rvm_error(log, "Cell did not new-copy/compare");
            cpy.display("Copy: ");
        }

        rvm_trace(log, "Copying cell to existing instance...");
        cpy = new;
        void = cell.copy(cpy);
        if (!cpy.compare(cell, diff)) {
            rvm_error(log, "Cell did not copy/compare");
        }
    }
}
```

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        cpy.display("Copy: ");
    }

    rvm_trace(log, "Packing/unpacking cell...");
    n = cell.byte_pack(bytes);
    if (n != bytes.size()) {
        rvm_error(log, psprintf("Reported number of bytes packed (%0d) differs from actual
                                n, bytes.size()));
    }
    if (n != cell.byte_size()) {
        rvm_error(log, psprintf("Reported number of bytes packed (%0d) differs from predict
                                n, cell.byte_size()));
    }
    m = cpy.byte_unpack(bytes);
    if (m != n) {
        rvm_error(log, psprintf("Reported number of bytes unpacked (%0d) differs from numbe
                                m, n));
    }

    if (!cpy.compare(cell, diff)) {
        rvm_error(log, "cell did not pack/unpack/compare");
        cpy.display("Unpacked: ");
    }
    rvm_trace(log, "Saving/loading cellame...");
    fp = fopen("cell.sav", "w");
    if (fp == 0) {
        rvm_fatal(log, "Could not open save file");
    }
    cell.save(fp);
    fclose(fp);

    fp = fopen("cell.sav", "r");
    if (fp == 0) {
        rvm_fatal(log, "Could not re-open save file");
    }
    cpy = new;
    if (!cpy.load(fp)) {
        rvm_error(log, "Unable to load cell");
    }
    fclose(fp);

    if (!cpy.compare(cell, diff)) {

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        rvm_error(log, "Cell did not save/load/compare");
        cpy.display("Load: ");
    }
}
log.report();
}
```